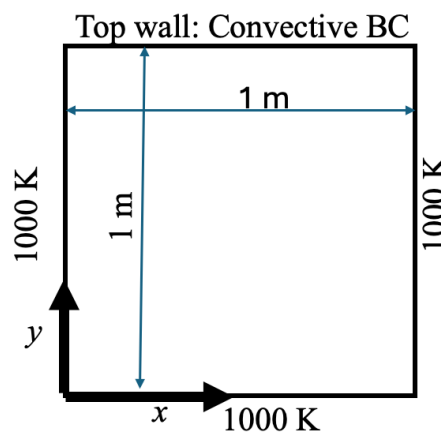


1. Consider the steady state heat transfer through a rectangular plate of size as mentioned in the figure. The left, right and bottom walls are at 1000 K, and the top wall is open to the surroundings. The convective boundary condition is applicable at the top boundary, where the heat from the wall is lost to the surroundings. The convective heat transfer coefficient $h = 10 \text{ W/m}^2\text{K}$, the surrounding temperature is $T_o = 300 \text{ K}$, and the thermal conductivity of the plate material is $k = 50 \text{ W/m.K}$. Find the steady-state temperature distribution inside the plate and plot the temperature contour using
 - a) Point Gauss-Seidel Method and plot the temperature distribution at $x = 0.0 \text{ m}$, 0.25 m and 0.5 m along y by extracting the data from the simulation.
 - b) Point successive overrelaxation (PSOR) (overrelaxation with the Point Gauss-Seidel method) and find the optimal value of the relaxation factor ω by plotting the number of iterations with respect to ω . You may consider a step size of 0.1 for the variation of ω .

Consider grid of 400x400 for all the methods. Finally, plot and compare the least number of iterations required in each method for a maximum permissible error of 0.1 K at a particular cell.



You are free to use any plotting tool for plotting the temperature contour and other graphs. Please prepare a report on your understanding and analysis, and submit the report along with the code.